particular negative potential V_0, called stopping or cut-off potential.

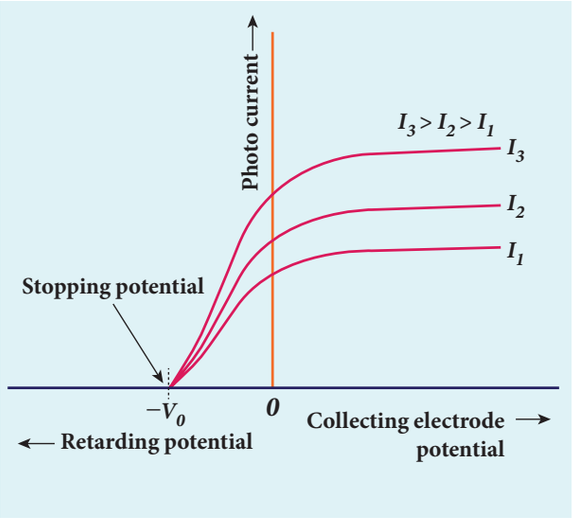


Figure 8.10 Variation of photocurrent with potential difference

Stopping potential is that value of the negative (retarding) potential given to the collecting electrode A which is just sufficient to stop the most energetic photoelectrons emitted and make the photocurrent zero.

At the stopping potential, even the most energetic electron is brought to rest. Therefore, the initial kinetic energy of the fastest electron (K_{max}) is equal to the work done by the stopping potential to stop it (eV_0).

$$K_{max} = \frac{1}{2} m v_{max}^2 = eV_0 \qquad (8.1)$$

where v_{max} is the maximum speed of the emitted photoelectron.

$$v_{max} = \sqrt{\frac{2eV_0}{m}}$$

$$v_{max} = \sqrt{\frac{2 \times 1.602 \times 10^{-19}}{9.1 \times 10^{-31}} \times V_0}$$

$$= 5.93 \times 10^5 \sqrt{V_0} \qquad (8.2)$$

From equation (8.1),

$$K_{max} = eV_0 \text{ (in joule)} \qquad \text{(or)} \qquad (8.3)$$

$$K_{max} = V_0 \text{ (in eV)} \qquad (8.4)$$

From the Figure 8.10, when the intensity of the incident light alone is increased, the saturation current also increases but the value of V_0 remains constant.

Thus, for a given frequency of the incident light, the stopping potential is independent of intensity of the incident light. This also implies that *the maximum kinetic energy of the photoelectrons is independent of intensity of the incident light.*

8.2.4 Effect of frequency of incident light on stopping potential

To study the effect of frequency of incident light on stopping potential, the intensity of the incident light is kept constant. The variation of photocurrent with the collecting electrode potential is studied for radiations of different frequencies and a graph drawn between them is shown in Figure 8.11. From the graph, it is clear that stopping potential vary over different frequencies of incident light.

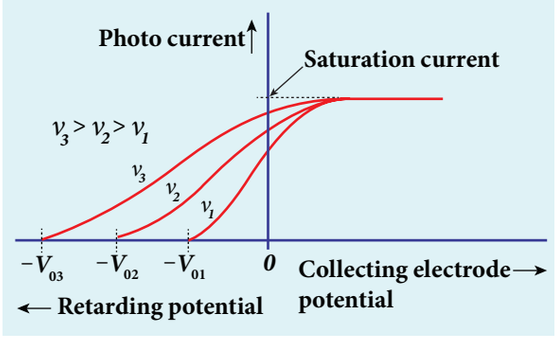


Figure 8.11 Variation of photocurrent with collector electrode potential for different frequencies of the incident radiation

XII_Physics_Vol-2_UNIT-8_EM.indd 115 03-03-2022 15:20:24